



Review

A review on security challenges of wireless communications in disaster emergency response and crisis management situations



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ABSTRACT

In all disaster management mission areas such as prevention, protection, mitigation, response and recovery, the first pressing need is to establish a communication network for data collection, and especially for reliable real-time information exchange between emergency actors. Generally, when a crisis happens all the pre-existing communication networks are down and the different responders have heterogeneous equipment. Because of the hurry of the situation, a rapid deployment of a reliable, easily configurable, robust, inter-operable, low cost and secure network is needed. Wireless networks provide a promising communication infrastructure to all the situation tiers such as headquarter and rescue operators. The security considerations are vital due to the sensitive and real-time need for information exchange, in the spatial context of chaotic emergency and crisis situations. Numerous security threats can be found in the different wireless based communication architectures proposed for crisis management. In this paper, we study disaster management security needs and we present an overview of the communication architecture proposed for emergency situations. We also provide a security analysis of these architectures and open issues to be tackled.

1. Introduction

The first thing to worry about in natural crisis, man-made disasters and emergency situations is to establish an efficient emergency process management. This process must be fast and secure due the emergency of the situation and to the sensitivity of the information exchanged.

The process of managing these situations is generally hierarchical (Giddens, 1984) but may be self-organised (Rose et al., 2005), this duality is discussed in Hartikainen and Harnesk (2009).

In real life, despite the every day emergency like cars accident and fires that need responses from the same organisations, big scale disasters and crisis involve the coordination efforts between public institutions and many different organisations. This cooperation is only possible with reliable communication. Generally in these crisis situations all the pre-existing communication networks are down and the different responders have heterogeneous equipments that need to establish a rapid deployment of reliable, easily configurable, robust, inter-operable, low cost and secure networks.

The management of emergency situations presents four different phases (Oberg et al., 2011):

- **Preparedness:** Preparation in terms of planning, training of people, updating of solutions on the basis of experience and information learned during the different phases of past emergencies.
- **Response:** Alert to a catastrophic event, activation of a control center for appropriate intervention on the territory, intervention of first responders.
- **Recovery:** Completion of rescue actions and reestablishment of normal conditions.
- **Mitigation:** of consequences through the implementation of measures that prevent other disasters and secure the disaster area.

An emergency situation refer to a situation of public emergency, provoked by natural or man-made catastrophes (e.g., fire, earthquake, flood, explosion, terrorist attack, big accident ...). First responders have to be deployed. These rescuers are organized in teams, with different transportation means, equipped with tools for the intervention (e.g., water or stretchers). The deployed teams may belong to one or more emergency

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Table 1
Analysis of security threats in the proposed solutions for Emergency Management.

Solutions	Confidentiality	Integrity	availability	Authentication	Resilience	Detection	Response
E-SPONDER (Casoni and Paganelli, 2011; Calarco et al., 2010)	X	X	X	X	X	X	
Dervin et al. (2009)	X	X		X			
WISECOM (Beriole et al., 2011a)	X	X		X			
SALICE (Del Re et al., 2013)	X	X		X			
Peng et al. (2009)							
Portmann et al. (Portmann and Pirzada, 2008)	X	X		X			
GeoBIPS (Bouckaert et al., 2006)	X	X		X			
Shibata et al. (2008)	X	X		X			
Pirzada et al. (2006)	X	X		X			
Calarco et al. (2010)	X	X	X	X	X	X	X
Chen et al. (2009)	X	X		X			
Iapichino et al. (2008b)	X	X		X			
UrgentMesh (Hadhrami et al., 2011; Al Hadhrami et al., 2012)	X	X		X			
Chiti et al. (Fantacci et al., 2010; Bartoli et al., 2013)	X	X		X			
Bai et al. (2010)	X			X			

response organisations (e.g., police, fire service, and emergency medical services). The existing infrastructure-focused networking do not give the minimum communication tools required to meet the challenges.

A major nature's fury or just a short circuit caused by transformer break can result in a significant or total loss in telecommunication services. In such situations, the emergency personnel will need to provide their own communication network.

The challenge is to maintain the network survivability in this particular hostile environment. A lot of events impact the normal operation of the wireless network (Sterbenz et al., 2010), and several threats may exploit a vulnerability of the network. Malicious attacks in large-scale disasters and environmental challenges can take advantage from the weakly connected channels, mobility of nodes or unpredictably long delays of the wireless network (Sterbenz et al., 2002; Ellison et al., 1997).

In this paper, we conduct a state-of-the-art analysis of the different security threats when using wireless communications in emergency and crisis situations. We classify the proposed solutions of the literature in different emergency service life cycles, such as preparedness, deployment, maintaining, extension and monitoring.

Other surveys are proposed in the literature. They focus mostly on the well known security aspects in the deployment and the use of the wireless communication networks in emergency and crisis situations, with little regard to their foundational underpinning (Channa and Ahmed, 2010; Li, 2011; Loukas et al., 2013a; Loukas et al., 2013b). These surveys mainly fall into two categories: light security analysis or restrictive security analysis. For example, Chana et al. (Channa and Ahmed, 2010) give a light description of a set of threats against wireless communications but do not consider threats that are specific to emergency situations. They focus mostly on technology-related definitions of the different standards and products.

In (Li, 2011) Yan Li presents in detail several score-based reputation techniques used in crisis management. In (Loukas et al., 2013a, 2013b) Loukas et al. present a taxonomy of an identified set of cyber attacks in emergency situations. They present a review of defense approaches of the identified cyber threats in different emergency management approaches. In Al-Janabi et al. (2017a) the authors are reviewing the research done on key agreements aspects in Body Area Networks emergency management. Their contribution is a comprehensive list of security requirements and attacks on the key agreement schemes Body Area Networks emergency management by giving an overview of methods used for security and performance evaluations.

Our survey takes a different approach in two major ways. First, we investigate the different security challenges in different proposed solutions from a general security point of view, relating each threat to the crisis situation. Second, we classify the existent solutions in different kind of network technologies to analyse the specific network security

threats of the proposed solution. Finally we focus on the deployment and maintenance of the communication network security functionalities that are provided in different existent solutions. A conclusion is given and future research trends are presented.

2. Background and motivation

The objective of this work is to analyse the current state of security in wireless communication networks used for emergency situations and crisis response management. In more detail, in this paper, we focus on security in different wireless networks technologies such WSNs, MANETs, mesh networks, cellular networks, satellites networks and VANETs. These types of networks are excellent candidates for emergency situations because of their simplicity of use and their rapid and, easy deployment, their auto-configuration and also the low cost of the correspondent equipments.

Nowadays, there are numerous studies and proposals that deal with the use of wireless network communication in emergency situations. However, these solutions do not explicitly consider the emergency situations' specific security requirements as we do here. In the literature, the proposed approaches focus on the study of security threats. However, most of them treat a specific network architecture, and do not follow general parameters. Our first step is to explore all the proposed frameworks in different wireless networks in order to find the weakness in security concerns and discuss the eventual solutions.

Real effective emergency management and crisis response require timely and coordinated decision making. Recent natural crisis and other catastrophic events demonstrate the need for strong communication networks, to provide effective information sharing and decision support. Security leaks in a communication medium affect the other emergency management efforts. In fact, a security threat can even affect the use of responder communication devices, where the decisions during an emergency must to be taken and transmitted reliably. The integrity of the orders and directions has an immediate effect on the efficiency of the intervention, while a denial of service threat suspend communication between the headquarter and first responders.

To help leaders making critical decisions information visualization and inventories are needed, we also must have collaboration tools for situational awareness and secure communication network.

The aims of establishing a communication network are as follows (Goss, 2001):

1. Enable multiple agencies to work together and to collaborate their efforts to resolve an emergency situation,
2. Provide a reachable data architecture that allows a wide range of public and private information sources to be integrated together to provide a comprehensive situational awareness, and

3. Provide a security communication model that allows emergency response agencies to share and access to their own sensitive information and inventories as well as the ability to share selective information with other participating agencies and first responders.

Security is also needed to allow information sharing based on agency-specific policies, location and temporal context, events, semantics and preferences. The incident reports from the agencies and public can be disseminated and shared according to the roles, incident types, locations and preferences (Degwekar et al., 2007).

3. General security requirements

Several initiatives to identify the threats against emergency situations and crisis management communication have been conducted over the past few years. In fact, the deployment and the use of wireless communication networks face many generic security threats and attacks (Wu et al., 2007; Wang et al., 2006; Naeem and Loo, 2009; Siddiqui and Hong, 2007; Redwan and Kim, 2008; Roy-Chowdhury et al., 2005). Reliable, secure and robust communication is crucial for successful emergency management operations (Channa and Ahmed, 2010; Portmann and Pirzada, 2008; Kose et al., 2012). During emergency situations, the deployed wireless communication network may be vulnerable to faults and security threats (Channa and Ahmed, 2010; Loukas et al., 2013a; Sakellari and Loukas, 2013). Wrong data can be created within the network by corrupted nodes especially during terrorist attacks. Confidential data or medical data of a victim can be stolen. Therefore, the security of the network should be maintained. In Tøndel et al. (2009), the authors identify two main types of attackers posing a threat to first responder in a disaster situation, namely news media and terrorists. News media launch passive attacks, in order to obtain tactical operation information. Terrorists can launch follow-up active attacks disrupting routing, forge communication to cross-out legitimate access to extend a physical terrorist attack.

The hierarchically structured nature of emergency operations, where information flows are upwards and decisions are downwards makes the trust between emergency entities crucial. Techniques based on reputation try to detect or predict any malicious or selfish behaviors and then publish or revoke them so as to enhance the overall security of the network. In Li (2011) the author examines in details several reputation score-based techniques, the problem of making decisions based on information given by flatter, dynamic, ad-hoc information sources, and how to transmit these orders to the other tiers organisations via wireless communication networks. To fulfill the rise of information security needs, many security challenges have been studied to establish and quantify the security of communications in different types of networks used in emergency situations, (Adam et al., 2008; Durresi et al., 2009; Karakostas and Markou, 2008; Papageorgiou et al., 2009).

Channa and Ahmed (2010) explores a comprehensive set of threats against wireless communications, but not threats specific to the emergency situations. Based on (Portmann and Pirzada, 2008) they reported that the security requirements for emergency communications include:

- Privacy: It ensures that the information is available to the authorised persons, and is achieved by applying cryptographic techniques.
- Data integrity: It maintains the completeness and accuracy of the transmitted information and ensures that no one has changed it on the way.
- Authentication: It is the process of verifying the identity of someone or something. It is ensured by key management: It is worthwhile to exchange security credentials (such as, keys, digital certificates) efficiently during emergency response communication.
- Access control: Normally, the access is provided depending on the identity, time, and location constraints.

In Michalas et al. (2012) the authors identify the communication patterns between three identified actors, an authority that initiates and

coordinates the emergency response, the first responders that intervene in the field and a third party that can be any information or service needed or supply the two first parties. In this paper, the authors presented SETS, a protocol based on the use of certifications to validate the legitimacy of the nodes. They also discuss the resistance of the protocol to replay, forced delay and man-in-the-middle attacks. The approach of using only certification authority special system for authentication was criticized earlier in Casoni and Paganelli (2011) where the system become unreachable in case of network disconnection. More specifically, Adam et al. (2008) and Casoni et al. (Casoni and Paganelli, 2011) present a careful analysis of the security issues and properties for an emergency network to find those that most affect the design of the whole crisis response network. Besides authentication, data integrity and access control, they include:

1. Data confidentiality and secrecy: Data confidentiality and secrecy aims to hide the content of a message to unauthorized users. It can be implemented with asymmetric or symmetric cryptography.
2. Data and origin authentication: Data and origin authentication is the capability of a system to identify the sender of a message correctly.
3. Authorization and accountability: Authorization is related to the granting of access rights to the resources of the system (or the network) to only specific users. It is usually performed after the authentication procedure, and is related to the accountability property.
4. Non repudiation: Non repudiation is related to the capability of a system to prevent a user from denying the sending (or the reception) of a given message.
5. Availability: This is the ability of a system to be prompt and usable when the user needs it. This property is particularly important in emergency communication networks.

The authors of Casoni and Paganelli (2011) and Adam et al. (2008) assume that the main security requirements for emergency communication networks are confidentiality and availability. In fact, confidentiality and availability will deny an attacker from exploiting the emergency situation to attempt an intrusion or from sniffing data for personal gain and perform a denial-of-service (DoS) attack to increase the impact of the disaster. The authors analyse authentication combined with authorization in order to reduce DoS attacks. They suggest to use certification authority authenticities in two levels of the network as a solution. A costly approach according to Loukas et al. (2013b).

A system with traditional key generating techniques such as using the feedback shift register is vulnerable to cypher attacks which render it to be ineffective and insecure. Ali (2013) suggest a novel method to overcome this problem using a Random Forest-Data Mining (RFDM) algorithm. It incorporates using new types of linear and nonlinear functions to mix the plain text with the key in the encryption process and the cipher text with the reverse key in the decryption process. The method uses two types of dual randomization to baffle the attackers.

A novel security framework to ensure that applications are trusted and secured against malicious and harmful application programs. This can be achieved by applying a new data mining technique called the Risk Filtering Data Mining algorithm (RFDM) (Patel et al., 2015) in order to enhance the quality of the risk analysis and to remove a major part of the false risks at runtime in the client-side mashup application program. This method allows quarantining/hibernating by prevention the exposure of valuable and sensitive information whether personal or otherwise into non-executable restrictive defined areas within the node program.

The authors in Al-Janabi et al. (2017b) prevent intrusions to the system by given a way to risk analysis of Intrusion Detection and Prevention Systems (IDPS) to an acceptable level while detecting the intrusions and maintaining effective security of a system.

Another solution to maintain the availability of the network and counter (DoS) attacks in an emergency MANET is proposed in Sterbenz et al. (2010). The authors propose the use of a firewall and an intrusion detection system to counter the attackers. This solution is also criticized by Loukas et al. (2013b) because it requires powerful terminals rather than the mobile devices typically used by the emergency services. In Loukas et al. (2013b), the authors affirm that ensuring network availability in a wireless emergency network, where no alternative or redundant technologies exist, remains a significant open issue. The authors enumerate the known communication security threats and identify the proposed countermeasures that are very costly and inappropriate to the context of emergency management.

4. Existing approaches to emergency management

In this section, we review existing approaches of emergency situations and crisis management. We classify them according to the network technology: Satellite-based, Mesh- based and hybrid-based networks.

4.1. Satellite based solutions

Satellite communications are widely used to establish a connection in disaster sites (S.-S. Working Group et al., 2008; Kose et al., 2012). They provide robust, reliable, flexible, quickly deployed, broadcasting capable and high-speed connections. They also provide a wide variety of services accessible from any locations at any time. The Digital Video Broadcast-Satellite (DVB-S, including DVB-S2) standard (E. ETSI, 2009) is widely employed due to its moderate equipments especially in its portable versions.

In Calarco et al. (2010), Wodczak (2012a), Wodczak (2012b), Paganelli et al. and Casoni and Paganelli (2011) the authors present a satellite-based system for wide area crisis scenario management called E-SPONDER. A project co-funded by the European Commission under its Security Program. Satellite communications play a vital part in the architecture of the project. E-SPONDER uses a communication backbone, relaying communication from the first responders units and the locally mobile operating centers or remote emergency operation centers. The project uses the ETSI DVB-RCS standard (Morello and Mignone, 2006) for providing a full-duplex satellite link between the

different units that allow 2 Mbps for the uplink (Skinnemoen et al., 2004) and a data rate limited to 10 Mbps for the downlink (Morello and Mignone, 2006). Fig. 1 illustrates the architecture of E-SPONDER project.

The DVB-S platform is complementary to a IEEE 802.11 wireless mesh network that provides a connection in the field within the rescue team. This wireless mesh network is used also to solve the problem of limited coverage of the mobile satellite service in indoor places. In fact, mobile satellite communications requires a Line-of-Sight (LoS) connection, which is impossible to maintain, especially when disaster teams work indoor areas. An Internet Protocol (IP) based application for maintaining flexibility and inter-operability complete the installation. The E-SPONDER project aims to give a secure communication infrastructure by using a two level of CA (Certification Authority) authentication in the network as a solution for reducing the DOS attack and for reinforcing authentication combined with authorization. The project proposes to store and distribute certificates among the users themselves to solve the lack of central authority or third party for secure key management. They also propose to use a distributed authentication scheme because the centralized RADIUS server scheme is not scalable in commercial WLAN systems.

Based on transparent satellite telecommunication, Dervin et al. (2009) propose an easy-to-deploy emergency communication model. The model relies on the underlay transmission of low-power emergency signals in the frequency band of a primary transparent satellite telecommunication system. Wide band spreading (Dervin et al., 2008) is used so as to guarantee that the primary system performance is not affected by the inter-system interference. This satellite based model is combined with heterogeneous networks depending on the land emergency terminal characteristics. It proposes various emergency communication services such as alert missions and voice communications. The use of several satellite transponders allows a rapid and simple deployment of a wide range of emergency services over communication networks connected with low development cost, provided by satellite in-use, with proper coverage and characteristics. The model uses a very wide band to superpose low power emergency transmission, on top of another satellite system in the same frequency (in Ku-band (Miranda et al., 1998)) to increase efficiency. In order to ensure that the primary system is not impacted by interferences with the emergency response signal, the solution provides a transition with data rates from 10 to 100 Kbps by the use of several satellite transponders without interactiv-

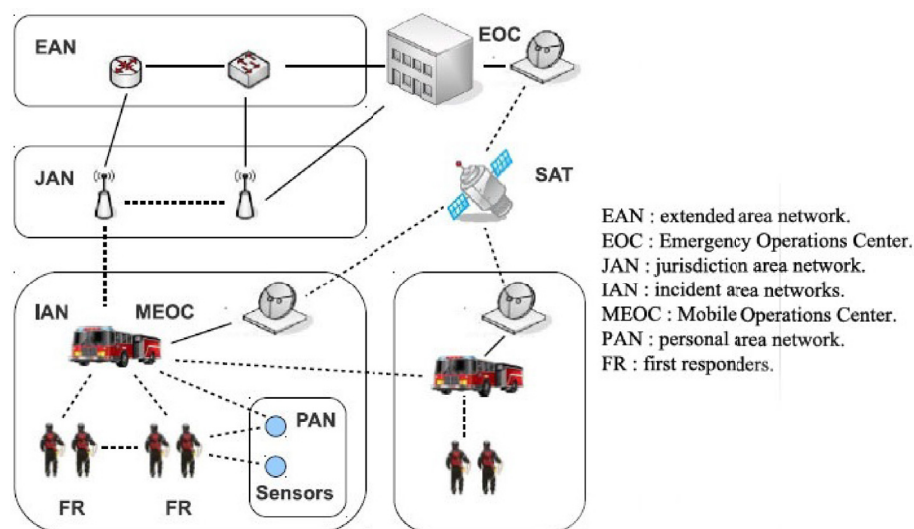


Fig. 1. E-SPONDER project architecture (Calarco et al., 2010).

ity. This ensures a good time availability over a widespread emergency area.

Dervin et al.'s (2009) model presents a low cost communication network based on transparent satellite. The authors present a study of the communication performance through the analysis of a two scenarios to avoid any impact on the primary system transmission performance. However, data transition realisability and user authentication policies are needed for securing the end to end information exchange.

In Skinnemoen et al. (2007), Thomasson et al. (2008) and Beriooli et al. (2007a, 2007b, 2011a, 2011b) the WISECOM European project is described. This project develops and validates a rapidly deployable lightweight communication infrastructure for emergency conditions (after a natural or industrial disaster). The system integrates terrestrial mobile radio networks - comprising GSM, UMTS, WiFi, and optionally WiMax and TETRA - over satellite and use Inmarsat BGAN and DVB-RCS systems. WISECOM uses lightweight and rapidly deployable technologies and incorporates location-based services. The infrastructure is intended to cover immediate needs in the first hours as well as medium to longer term needs during the recovery and rebuilding phase following an emergency. Fig. 2 illustrates the architecture of WISECOM project. In this project, the authors focus their model on the problem of inter-operability between all participating networks and all necessary functions that must be made available to users in different networks. They propose the use of a distributed aerospace platform in wireless communication for emergency rescue applications that focus on the integration of the aerospace segment with terrestrial backbones and ad hoc terrestrial networks for both data connections and assisted localization. They interline the benefits stemming from hybrid approaches using combinations of terrestrial and satellites networks. In fact, high altitude platforms (HAPs) can be used as emergency backbone for long-range connectivity and ad hoc wireless networks deployed on-site are, for sure, best suited to recover local connectivity in limited areas. In the context of cooperative strategies both at physical and at networking layers, an authentication and privacy policy is required to overcome the problem of interoperability between different networks to archive the exchange of information among the different networks.

Del Re et al. (2013) and Del Re (2011, 2012) describe the Italian National Research Project SALICE, which was funded by the Italian Ministry of University and Research in 2007. SALICE focused on two main aspects: 1) Achieving global coverage of the emergency area by developing a reconfigurable cooperative navigation and communication system. 2) Integrate heterogeneous communication systems for global coverage through the integration and cooperation of space (satellite and high altitude platforms (HAPs)) and terrestrial segments supporting available navigation and communication systems. The SALICE terminal (software-defined radio) based handset integrates navigation and communication capabilities. The project uses a Satellite/terrestrial cooperation: among the different phases of public emergency situation management (preparedness, response, recovery, and mitigation). This cooperation is performed through an anchor node, which exchanges satellite data traffic with other first responders in the a MANET over the Wi-Fi interface, using ODMRP a mesh-based routing protocol for MANET that guarantees high scalability and reliability (Viswanath et al., 2006).

In Peng et al. (2009) a description of a multimedia emergency command system based on satellite IP link is presented. The characteristics include an on-the-spot information collection and transmission, an integrated emergency communication, an emergency commanding, an emergency dispatching and strategic decision analysis. The aim is to allow emergency mobile command to enhance the efficiency of inter-connection establishment. The system can transmit video, audio and data information of geological disasters at any time with an Internet Protocol (IP) based satellite link for practical reasons and flexibility. Authentication is needed during connection establishing of different mobile commands to avoid that fake users have access to the commu-

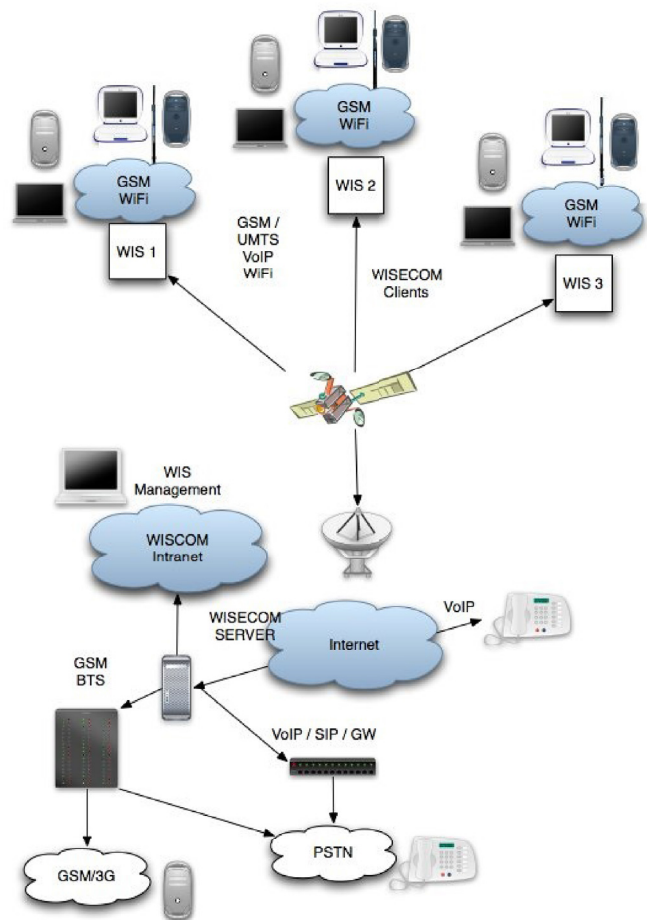


Fig. 2. WISECOM project architecture (Skinnemoen et al., 2007).

nication network. Privacy and data integrity are also needed to provide secrecy and accuracy of sensitive data.

4.2. Mesh network based solutions

Different wireless mesh networks have been proposed in disaster management because mesh networks give a promising communication technology for first responders. A wireless mesh network is a backbone based wireless multi-hop network with auto-configuring capabilities, rapid deployment and guarantees a high degree of robustness and fault tolerance. The three main kinds of wireless mesh network architectures (Infrastructure mesh, client mesh and hybrid mesh networks) are based on IEEE 802.11 technology (Kafaie et al., 2018; Portmann and Pirezada, 2008).

In Portmann and Pirezada (2008) the authors discuss the suitability of wireless mesh networks technology to facilitate communications among first responders during crisis management. They distinguish in (Miller) four hierarchical communication areas: personal area network (PAN) represented with mesh clients, incident area network (IAN) presented by mesh routers, jurisdiction area network (JAN) presented by mesh gateways and extended area network (EAN) the external network. These systems support information exchange among first responders (PAN) and emergency headquarters (EAN) during emergency management operations.

In Bouckaert et al. (2006) a mesh wireless network system, called GeoBIPS has been proposed to provide support for communication during crisis situations. GeoBIPS is a self-forming broadband wireless mesh

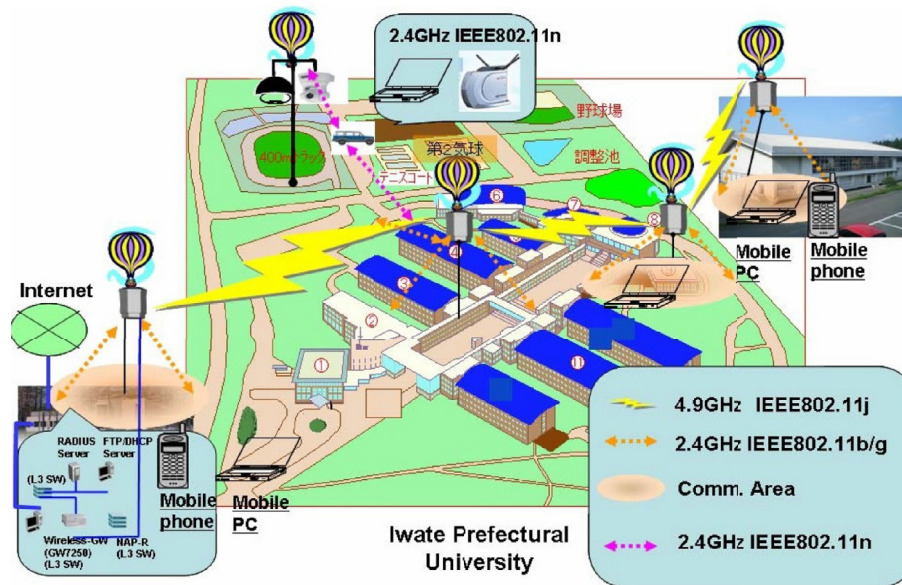


Fig. 3. Ballooned wireless mesh network for emergency information system (Shibata et al., 2008).

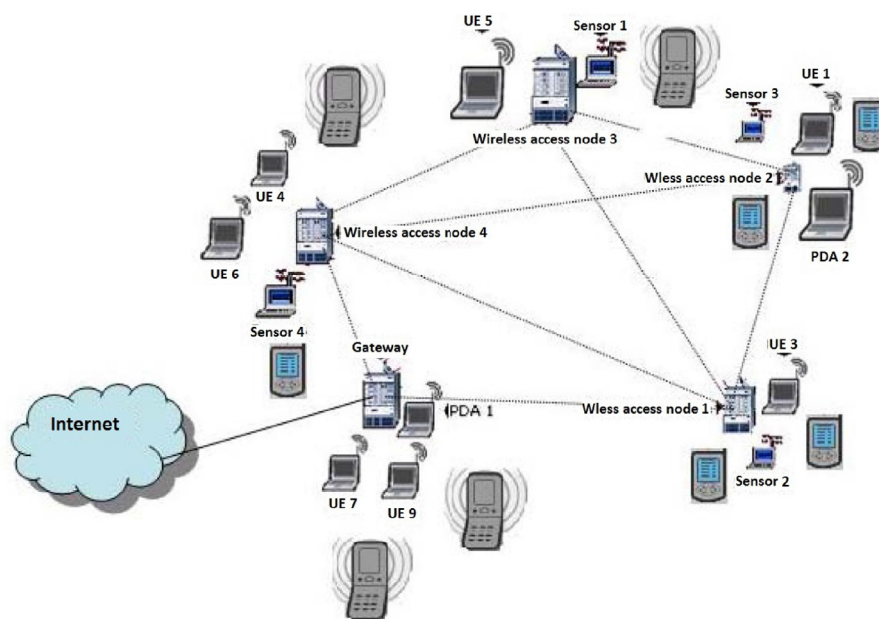


Fig. 4. A wireless mesh infrastructure deployment with application for emergency scenarios (Dilmaghani and Rao, 2008).

network used by first responder team members and commanding officers. The mesh network is used as relay interface between first responders, commanding officer and crisis center for decision making to send useful data and instructions. GeoBIPS's architecture extends the use of local hot-spots, general packet radio services (GPRS) or a universal mobile telephone system (UMTS) connection to communicate with a remote networks.

In Shibata et al. (2008) the authors propose a ballooned wireless mesh network to provide reliable communication means for exchanging information relating to disaster area. The system comprises of multiple ballooned wireless nodes. The meshed backbone hanged in the sky provides a large network coverage connection for first responder devices, such as fixed access point, mobile notebook PCs and wireless

IP phones. The network is connected to the Internet and disaster information server (DIS) through a fixed access points. Fig. 3 illustrates the architecture of this (Shibata et al., 2008).

In Pirzada et al. (2006) the authors study routing protocols effects in wireless mesh network proposed communication network for crises management. An algorithm for modifying the routing protocol AODV (Perkins and Royer, 1999) to route through 802.11 mesh networks has been studied on a test-bed. It results from that study that routing affects the quality of service (QoS) of the video applications between the rescue teams and the command center. For this reason, the authors intend to further investigate routing algorithms in further work.

In Calarco et al. (2010) the authors show the successful use of wireless mesh networks in emergency response situations by verifying it in

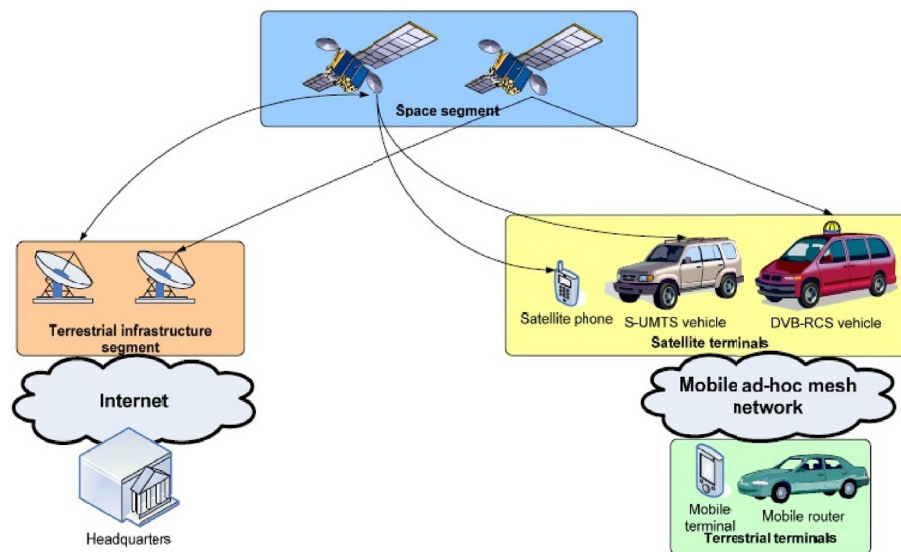


Fig. 5. Advanced hybrid satellite and terrestrial system architecture (Iapichino et al., 2008a).

real crisis situations such as hurricane Katrina and the tsunami of 2004. They present a complete analysis of the features and requirements of wireless mesh nodes supporting visual communication during emergencies and natural disasters such as high-quality video, image with high resolution, deployment, operation, battery life, maintenance, configuration, capacity, security, monitoring and cost.

The authors in Chen et al. (2009) developed a hybrid wireless mesh network topology to be deployed in earthquake situation management. A general hierarchical topology is presented in Dilmaghani and Rao (2008). It is used for general emergency situations and natural disasters, not just for a special case such as earthquakes. Fig. 4 illustrates the deployment of the approach described in Chen et al. (2009). The advantages of a hybrid wireless mesh network topology are shown to be quick deployment, easy configuration and inter-operability.

In Hadhrami et al. (2011) a routing metric is proposed to justify the use the OSPF routing protocol (Moy, 1989) for cost-effective integration the wireless mesh networks and internet, to simplify communications between the incident area and the remote headquarters e.g. via satellite links. In Hadhrami et al. (2011) the authors propose the use of Power- and Node-Type-Aware Routing Algorithm (PNTARA) for optimising route selection between two arbitrary wireless mesh networks nodes and the mesh router backbone network deployed in the incident area. PNTARA is proposed in Al Hadhrami et al. (2013) to select optimized routes based on a joint consideration of the nodes' types and power levels along the path.

4.3. Hybrid wireless networking based solutions

In Iapichino et al. (2008a, 2008b) and Iapichino and Bonnet (2009) the authors present a heterogeneous communication infrastructure that cover the full mobility of rescue teams and the covering of bi-directional communication needs for voice and data in the first critical hours following an emergency. This is a hybrid IPv6-based architecture that integrates hybrid satellite and wireless terrestrial networks. Vehicle Communication Gateways (VCGs) create an inter-vehicular mobile ad-hoc mesh IEEE 802.11s network in the emergency field providing connectivity to isolated IPv6 cells and connects vehicles via DVB-RCS. The vehicles are equipped with a very compact and transportable satellite terminal to the disaster area with the headquarters, as shown in the Fig. 5. A combination of Proxy Mobile IPv6 (PMIPv6) (Gundavelli et al., 2008) and Host Identity Protocol (HIP) (Moskowitz et al., 2008)

using an Authentication, Authorization and Accounting (AAA) server. The (AAA) server provides to each Mobile Node an identifier, to helps rescue teams to easily move and keep their connections on while moving under different mobile routers and switching from one access technology to another and establishing security connections with peers. The confidentiality and data integrity is assured by the use of the IPSec Encapsulated Security Payload (ESP) and Security Association (SA) pair between endpoints using a Diffie-Hellman authenticated key.

In Hadhrami et al. (2011) and Al Hadhrami et al. (2012) UrgentMesh system is proposed based on the use of wireless mesh networks and Digital Video Broadcast-Satellite (DVB-S) system to provide an effective way of delivering high-quality video between the head office and the rescue team in the field. Security is assured by the use of a self-configuring and secure infrastructure mesh network architecture called Mesh Cluster designed and implemented in E-SPONDER (Casoni and Paganelli, 2011) to assure authentication and privacy in the wireless mesh segment. Fig. 6 illustrates the architecture of this solution (Hadhrami et al., 2011; Al Hadhrami et al., 2012).

In Fantacci et al. (2010) and Bartoli et al. (2013) a Broadband Wireless Network system is presented, this WiFi-WiMAX based solution uses a WiMAX backbone for integrating WSNs (monitoring sensitive areas) and connecting heterogeneous networks, or as communication system providing ubiquitous mobile services based either on a point-to-multipoint or mesh architecture. Availability and robustness of the network are provided by the use of security services such as privacy, authentication and access control, based on digital certificates. The authentication is based on the presence of a centralized authentication server, reachable with the Extensible Authentication Protocol (EAP) and the Remote Authentication Dial-In User Service (RADIUS) protocol. An ad hoc solution is used for the wireless sensors network segment based on the knowledge of the cryptography keys and on the localization of the nodes. The solution also considers intrusion sensing and behavior analysis of the nodes forming the network to identifies suspicious behaviors. Fig. 7 illustrates the architecture of this solution (Fantacci et al., 2010) in Bartoli et al. (2013).

In Bai et al. (2010) the authors propose an integrated heterogeneous communication system. They deploy in first time a wireless sensor network (WSN), a wireless mesh network and a mobile ad hoc network on the disaster site for local communication and information collection. They use a satellite gateway to establish communication with the headquarter. A cellular gateway is used too as an alternative remote

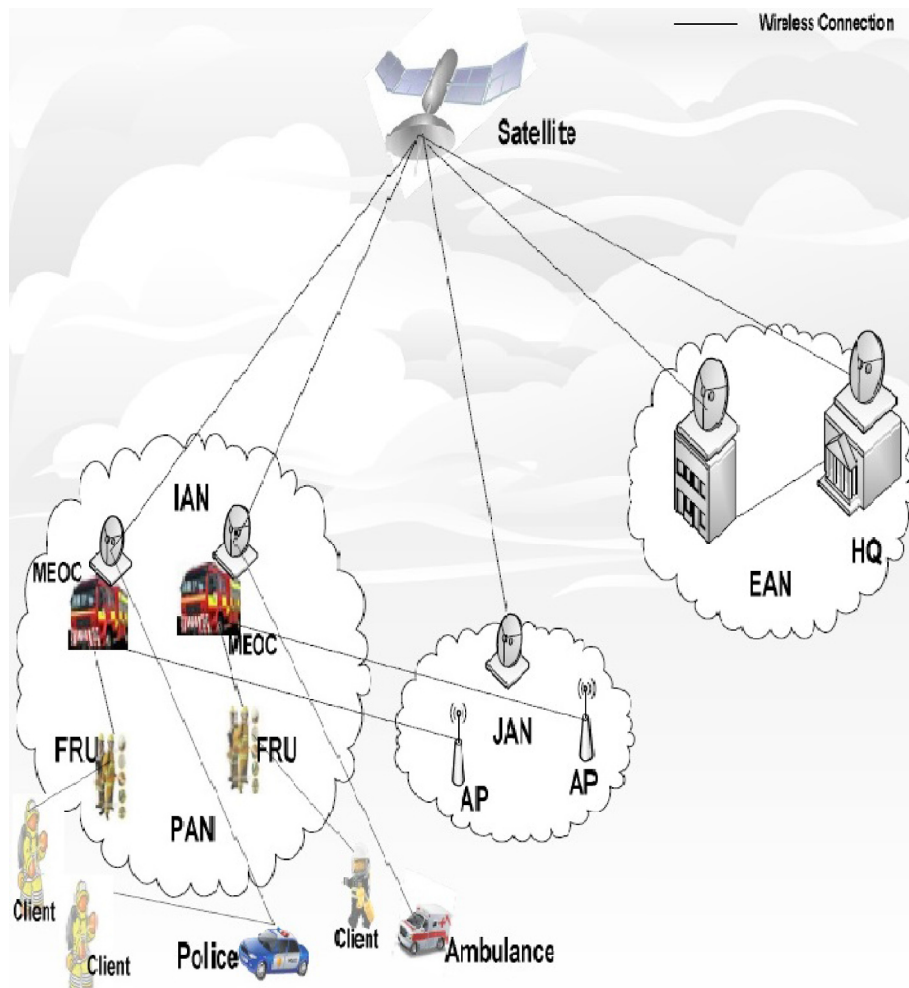


Fig. 6. UrgentMesh architecture (Hadhrami et al., 2011).

communication means when the expansion of local networks reaches a working cellular base station. The communication paths are chosen according to the locations and the wireless network coverage conditions. A security scheme service must be used to secure the end to end communication between the different devices operating within the heterogeneous network systems using different security stack to avoid the unnecessary security redundancy. Fig. 8 illustrates the architecture of this solution (Bai et al., 2010).

5. Discussion

Availability and robustness of the communication network are fundamental for any catastrophes and crisis situations. The security services must be provided for reliable and secure communication through privacy, authentication, or access control. Our study of the majority of the architectures proposed in the literature for disaster management shows that these solutions either don't give any security answer or don't give an adequate security service. Three Main challengers must be taken in consideration. In fact, the distributed nature, the heterogeneity of the networks and the requirement of availability of real-time communications, make a real challenge in the security of the proposed communications architecture.

1. Centralization: The highly dynamism of the rescuers in emergency situations need the use of distributed networks for their flexibility and auto-reconfiguration. However, an efficient distribution

of the responsibility is needed to address the security weakness of the most authentication and access control protocols, where a centralized authority is often used (ex: centralized authentication server not always reachable). Most of wireless-mesh network based architecture use IEEE 802.11i, a security standard to provide privacy, integrity, and authentication at the MAC layer of IEEE 802.11 networks. IPSec and TLS protocols are used in IP and transport layers. However, the key management distributing across multiple agencies implied in disaster management is a big deal to ensure the reality of the network. In fact, the use of pre-shared keys or certification authority for key distribution do not hold during disasters situations when emergency service units from various agencies and jurisdictions must communicate.

2. The heterogeneity: The heterogeneity of devices carried by rescuers and wireless networks used in the communication network must be addressed. The devices that usually operate within a disaster area are different, both in terms of the protocols used and the hardware capacity. Despite this, end-to-end communications between entities involved in the rescue operations must be secure. As an example, we can consider the case of data initially acquired by a sensor node, successively conveyed along a multihop sensor network, then routed to a WiMAX access point, and finally delivered to a remote control center using the WiMAX backbone network. Although IEEE 802 network security is based on digital certificates, this technique is not applicable to wireless sensor network devices due to their lim-

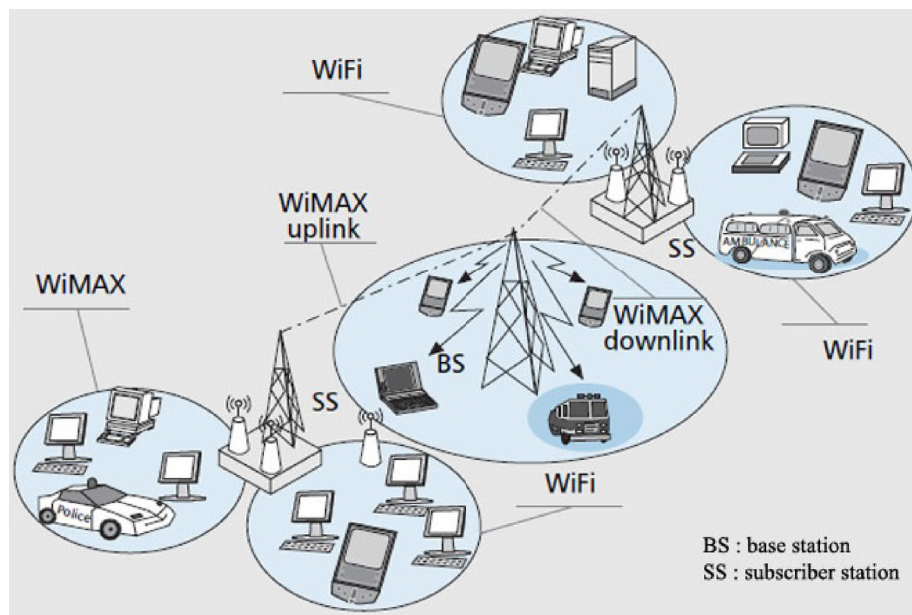


Fig. 7. Heterogeneous WiFi-WiMAX network broadband architecture (Fantacci et al., 2010).

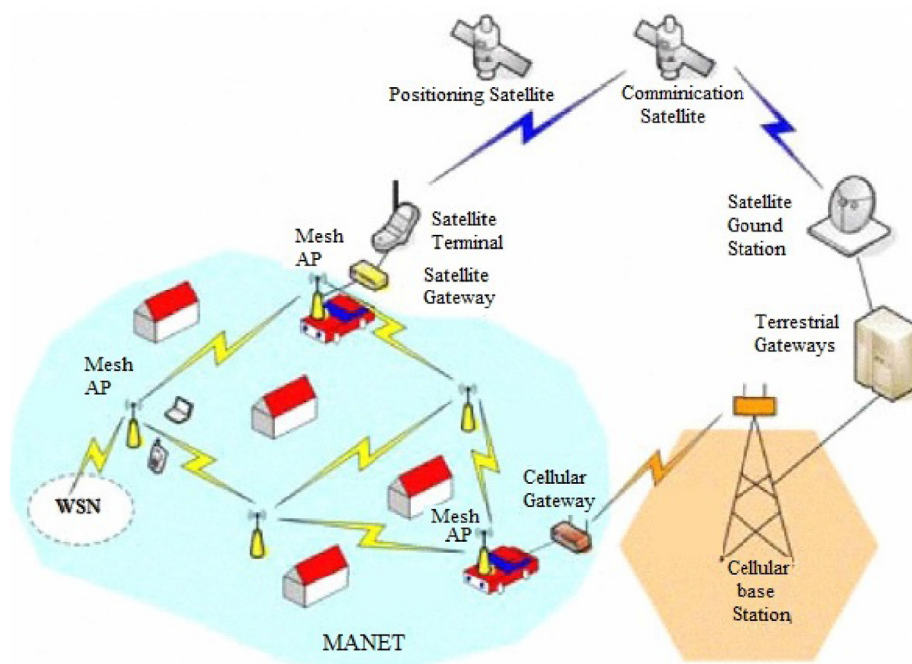


Fig. 8. Diagram of integrated emergency communication system (Bai et al., 2010).

ited hardware capability. So, the heterogeneity of the system does not permit the use of a single security method. However, each of these networks contain its specific security services, such as access control or data authentication. Even if the single algorithms may be different, the security services are generally the same. So, the definition of proper inter-working system (IWS) functions between the security stacks of each communication media can help interoperability. IWS functions are required so that the information coming from a WSN using certain security services are matched with equivalent services on the backbone network using a different security stack.

3. Overhead: Emergency communication systems often transport data with stringent time constraints, such as voice, video, or software calls to remote systems. Security protocols introduce an overhead in terms of packets exchanged and computational complexity that can conflict with these constraints. As an example, IEEE 802.11/WiMAX based networks require a complete re-authentication of the terminal with the centralized authentication server when performing handovers. This procedure might involve the exchange of several packets and verification of digital signatures. So, it gives rise to significant service degradation in terms of delivering delay. This situation is made even worse for vertical handovers, when security proce-

dures must be bridged between multiple subnetworks. As two practical examples of challenges introduced by security requirements, we describe two real case open issues and solutions suggested in the literature. The first scenario is represented by IEEE 802.11 wireless mesh networks: in these kinds of networks, the authentication is based on the presence of a centralized authentication server, reachable with the Extensible Authentication Protocol (EAP) and the Remote Authentication Dial-In User Service (RADIUS) protocol. Whatever the authentication method which is applied (shared key, transport layer security [TLS], etc.), at least a four-way handshake between the roaming node and the server is required. This overhead makes the network non-scalable when the number of nodes increase. Moreover computationally heavyweight authentication protocols, such as TLS, can overload a single server. In [Cronstedt \(2002\)](#), the problem was analyzed, and a solution was proposed based on the reduction of the number of packets required for re-authentication opposed to the first complete authentication, which is applicable in realistic networks. The second issue is key-establishment and eventually, authentication in WSNs. This field is very active and has produced interesting solutions, because the lack of hardware capabilities, the proposed solutions use low-cost crypto primitives. In such a field, original schemes for pre-distribution keys or the use of functions that can approximate the behavior of public/private key schemes (such as multivariate polynomials) are under active research, although far from standardization. WSNs are still perceived as dedicated networks where ad hoc solutions are applied to each specific need, and no generic security procedure is present now.

[Table 1](#) shows an analysis of security threats in the reviewed solutions for Emergency Management.

Ensuring network availability in a wireless emergency management situation, where no alternative or redundant technologies exist, remains a significant open issue. Relevant research on the availability specifically of emergency opportunistic communications (Oppcomms) has been carried out by [Gelenbe et al. \(2012\)](#). Oppcomms are a type of delay/disruption-tolerant network where communication nodes store messages on behalf of others and forward them to other communication nodes when they meet, they investigate the impact of cyber attacks in the cyber-physical-human ([Madni et al., 2018](#)) context of a technology-assisted evacuation. They provide a communication infrastructure in situations where connectivity is intermittent. This makes them naturally resilient to availability attacks. Nevertheless, cyber attacks can affect their performance considerably and severely handicap the emergency operation. The results shown that even a single attacker can have significant impact on an evacuation that is heavily dependent on communications. To tackle this issue, they have proposed identity-based signatures and content-based message verification to block malicious traffic and nodes that could disrupt communications during a building evacuation. The mechanism is collaborative, with network nodes communicating with each other to establish the consistency of emergency information ([Gorbil and Gelenbe, 2012, 2013](#)).

6. Conclusion

In this paper we present a study of security challenges in emergency situations and review existing communication proposals for emergency response and their security solutions. In these scenarios an attack can either be used by opportunistic persons to steal data or money exploiting the confusion and hurry of the situation or by terrorists to complete a physical attack or spread fear and confusion.

Our studies conclude that since so far there is no system that provides all security services in any network. Almost all studied schemes focus on authentication and access control to solve security problems, but this not enough. Some nodes might demonstrate malicious behavior by intercepting secret information. An attacker

in possession of a legitimate node has full access to the network and can conduct several types of attacks. Moreover, the joining of new nodes to the network, which is an inevitable step to prolong its life, presents the best opportunity to conduct such an attack. The existing schemes suffer from the absence of revocation mechanism of corrupted or compromised nodes or a reactive security mechanism to complete the security reliability to avoid mismanagement of emergency response efforts, where the readability of the communications is crucial and human lives depend on rapid and efficient intervention.

We have provided a review of security challenges in wireless communications in emergency situations and presented the lake in security of the existent communication framework using different wireless technologies, highlighting the open issues for research. We have attested that the main difficulty in providing a complete security coverage is identify the security threats of each phase of the emergency situation and give an appropriate security preventing and reactive solutions.

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